

COMPUTATIONAL APPENDIX: REVISION R30

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1. READING THE EVIDENCE

All exact fractions, candidate parameters, raw failures, region covers, derivative enclosures, and source identities are stored under `revisions/2026-09-24-r30/results`. Displayed decimal performance bounds are rounded outward. Timings are single-environment observations, not confidence intervals. The article and supplement give model and theorem scopes; this appendix does not extend them. The largest cost-enclosure gap is at most $2.28\text{E-}10$.

TABLE I
FROZEN NEURAL CANDIDATES: RAW ACCURACY AND EXACT PROOF OBJECTS

Gain	Raw passes	Median cells	Bound calls	Max bits	Certificate KiB
Linear	19/20	4	2–80	203	0.917–29.79
Cubic	0/20	112	83–286	624	47.83–132.6
Quadratic	7/20	77	6–152	415	3.056–73.23

Raw passes require an independently enclosed worst-state, all-restart regret no greater than .01. Counts are not independent economic replications. Certificate size includes exact rational endpoints and proof bounds.

TABLE II
OBSERVED CONSTRUCTION WORK IN SECONDS

Gain	Fit	Compile	Cert.+check	Total	Structural	Grid DP
Linear	0.04639	0.001201	0.0004445	0.04901	0.002272	0.2178
Cubic	0.06541	0.001625	0.0521	0.1156	0.01557	0.04955
Quadratic	0.06373	0.002	0.02132	0.08557	0.006683	0.03602

Each entry is descriptive single-environment timing. Total is the median of each candidate’s full sum, not a sum of separately rounded medians. Structural is the matched-tolerance Bernstein solver; it is not the exact symbolic sign rule. Grid DP materializes 2^{20} states and uses binary64. All methods have the same model primitives.

All three polynomial proposals meet the raw .01 criterion; their details are in the computational appendix. Among the neural comparisons with positive saved intervention cost, the largest break-even shadow price is below 0.000739. At the illustrative price $\lambda = 1/20$, 60 of 60 exact paired comparisons favor the completed policy over the near-exact structural policy after intervention cost. This is an accounting comparison, not a statistical replication count.

TABLE III
DISCOUNTED ECONOMIC INTERVENTION METRICS

Gain	Occupancy edits	Priority cost	Operating gain	Extra-cost upper
Linear	0	0	0	$4E - 12$
Cubic	2.382	3.783	0.2736	$2.28E - 10$
Quadratic	0.1387	0.1934	0.001208	$8.4E - 11$

First three numerical columns are medians across the 20 neural candidates. The last is an outward-rounded upper bound on the largest observed upper-minus-lower cost enclosure in the stated local-loss class. All quantities concern the continuous reset density.

TABLE IV
UNCHANGED ORIGINAL ECONOMY AT THE ACTIVE-BOUNDARY RESTART

θ	Exit probability	J_θ	J'_θ	J''_θ
-0.2	[0.99997, 0.99998]	[-7.6652, -7.6650]	[1.7017, 1.7019]	[16.9990, 16.9992]
-0.1	[0.98580, 0.98581]	[-7.3389, -7.3387]	[5.9044, 5.9046]	[82.1152, 82.1154]
0	[0.68915, 0.68916]	[-6.2320, -6.2318]	[15.9905, 15.9907]	[60.6513, 60.6515]
0.1	[0.19903, 0.19904]	[-4.6662, -4.6660]	[12.7887, 12.7889]	[-83.8317, -83.8315]
0.2	[0.04076, 0.04077]	[-3.7609, -3.7607]	[6.0310, 6.0312]	[-45.1336, -45.1334]

All endpoints are rounded outward from validated balls. This is one restart and a fixed constant-control class, not the original whole-domain target. The opposite-boundary and time-truncation errors are included.

TABLE V
THE SAME VALIDATED CENTRAL-STATE ORACLE UNDER TWO SEARCH PROCEDURES

k	Setup+audit s	Bracket calls	Poll calls	Bracket s	Poll s	Bracket bound	Poll bound
1/2	29.6	2	23	0.02666	0.4102	0	0.005
2	29.19	8	31	0.1074	0.5193	0.00371	0.007763

The bracketing target is .005. The inherited poll meshes have the achieved bound displayed here, not an artificially matched value. Both methods require the same oracle construction and concavity audit.

TABLE VI: All neural candidates; A = Adam, B = L-BFGS

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
linear	A/8/1	0.000039	3	3	0	0	0.9121
linear	A/8/2	0.012663	41	77	0.03773	0.0516	0.07108
linear	A/8/3	0.000024	3	3	0	0	0.06399
linear	A/8/4	0.000783	4	4	0	0	0.06383
linear	A/8/5	0.000007	4	4	0	0	0.06447
linear	B/8/1	0.000003	3	3	0	0	0.02075
linear	B/8/2	0.000009	5	5	0	0	0.0221
linear	B/8/3	0.000001	3	3	0	0	0.01558
linear	B/8/4	0.000006	2	2	0	0	0.008753
linear	B/8/5	0.000002	3	3	0	0	0.01174
linear	A/16/1	0.000167	4	4	0	0	0.06609
linear	A/16/2	0.004521	41	77	0.009016	0.01237	0.07411
linear	A/16/3	0.000006	2	2	0	0	0.06553
linear	A/16/4	0.000024	3	3	0	0	0.06516
linear	A/16/5	0.002148	44	80	0.0004114	0.0005666	0.07409
linear	B/16/1	0.000002	6	6	0	0	0.02501
linear	B/16/2	0.000025	6	6	0	0	0.02479
linear	B/16/3	0.000257	6	6	0	0	0.009412
linear	B/16/4	0.000008	5	5	0	0	0.01304
linear	B/16/5	0.000001	7	7	0	0	0.03418
cubic	A/8/1	0.420749	144	286	2.935	4.579	0.127
cubic	A/8/2	0.071483	115	223	0.2479	0.3731	0.1147
cubic	A/8/3	0.384982	110	217	2.387	3.79	0.113
cubic	A/8/4	0.384866	110	217	2.386	3.789	0.1169
cubic	A/8/5	0.420749	144	286	2.935	4.579	0.1844
cubic	B/8/1	0.420749	144	286	2.935	4.579	0.06817
cubic	B/8/2	0.030503	80	152	0.09845	0.1649	0.1278
cubic	B/8/3	0.385395	110	217	2.39	3.795	0.06425
cubic	B/8/4	0.385395	110	217	2.39	3.795	0.06486
cubic	B/8/5	0.385392	110	217	2.39	3.795	0.06136
cubic	A/16/1	0.384537	111	218	2.383	3.785	0.1165
cubic	A/16/2	0.403565	110	217	2.562	4.034	0.1108
cubic	A/16/3	0.384208	112	219	2.38	3.781	0.1208
cubic	A/16/4	0.045918	115	223	0.1029	0.1659	0.1229
cubic	A/16/5	0.018225	47	83	0.04472	0.08034	0.09018
cubic	B/16/1	0.032401	116	224	0.1247	0.1975	0.1729
cubic	B/16/2	0.032752	115	223	0.1425	0.2324	0.09924
cubic	B/16/3	0.383961	112	219	2.378	3.779	0.09589
cubic	B/16/4	0.032828	115	223	0.1442	0.2349	0.1377
cubic	B/16/5	0.029961	117	225	0.09242	0.1585	0.1801
quadratic	A/8/1	0.059549	73	144	0.8961	1.209	0.08238
quadratic	A/8/2	0.001982	6	6	0	0	0.06422
quadratic	A/8/3	0.059549	74	145	0.8961	1.209	0.08016
quadratic	A/8/4	0.058462	77	149	0.5674	0.7913	0.08809
quadratic	A/8/5	0.059549	74	146	0.8961	1.209	0.08306
quadratic	B/8/1	0.059549	73	144	0.8961	1.209	0.02752
quadratic	B/8/2	0.014738	77	149	0.1417	0.1945	0.05122
quadratic	B/8/3	0.010987	78	150	0.07094	0.104	0.0445
quadratic	B/8/4	0.015021	77	149	0.1403	0.1922	0.0612
quadratic	B/8/5	0.203274	74	146	0.9891	1.307	0.02883
quadratic	A/16/1	0.005034	78	150	0.02653	0.03837	0.08954
quadratic	A/16/2	0.059175	78	150	0.8797	1.186	0.08849
quadratic	A/16/3	0.046384	77	149	0.4816	0.6568	0.0917
quadratic	A/16/4	0.004746	78	150	0.01232	0.01552	0.09207

Gain	Method/width/seed	Raw regret upper	Cells	Calls	Occupancy	Priority	Full s
quadratic	A/16/5	0.024038	77	149	0.1023	0.1259	0.08828
quadratic	B/16/1	0.006289	79	151	0.03104	0.04194	0.134
quadratic	B/16/2	0.005631	79	151	0.02927	0.03989	0.1475
quadratic	B/16/3	0.005626	79	151	0.02757	0.0374	0.156
quadratic	B/16/4	0.021356	42	78	0.1371	0.2043	0.05929
quadratic	B/16/5	0.004861	80	152	0.02913	0.04057	0.2042

TABLE VII

POLYNOMIAL CONTROLS WITH THE SAME OBSERVATIONS AND KNOWN DEGREE

Gain	Raw regret upper	Fit s	Compile s	Certify s	Cells
linear	0E-12	0.002705	0.001906	0.000161	2
cubic	1E-12	0.0001478	0.01762	0.0008435	4
quadratic	1E-12	0.0002166	0.006903	0.0002222	3

These are exact compiled dyadic policies from fitted polynomial coefficients. An irrational root is not asserted to equal its dyadic approximation.

TABLE VIII
UNCHANGED R29 EXHAUSTIVE BENCHMARK AND TOTAL-WORK RATIOS

d/L	States	DP s	Repair+check s	Ratio	Full s	Full/DP
2/4	16	0.001072	0.003879	3.618	–	–
3/4	64	0.00803	0.01485	1.849	–	–
4/4	256	0.03729	0.0654	1.754	2.356	63.19
4/5	625	0.103	0.1841	1.787	3.464	33.63
5/4	1,024	0.2064	0.3054	1.48	4.729	22.91
6/4	4,096	1.103	1.484	1.345	14.81	13.43
7/4	16,384	5.495	8.345	1.519	59.95	10.91

Full includes new fitting, budget repair and verification only. Unmeasured or unavailable costs are not silently added as zero. These ratios use the displayed inherited stage times.